

Outputs

```
wordcount.py 2, M X input.txt M X
ITCS6190-H5-Hardi > wordcount.py > ...
1 """Word count in PySpark, with the same rules as the Hadoop job from Hands-on L4:
2 split on whitespace, ignore words shorter than the minimum length, sort by count.

Problems 2 Output Terminal Ports
zsh - ITCS6190-H5-Hardi

$withThreadLocalCaptured$1 at CompletableFuture.java:176 34 words scanned Aa ab .* 1 of 2
26/09/15 16:11:04 INFO TaskSchedulerImpl: Adding task set
26/09/15 16:11:04 INFO TaskSetManager: Starting task 0.0 in stage 11.0 (TID 6) (172.20.0.4,executor 0, partition 0, N
ODE_LOCAL, 9635 bytes)
26/09/15 16:11:04 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locations for shuffle 3 to 172.20.0.4
:55548
26/09/15 16:11:04 INFO TaskSetManager: Finished task 0.0 in stage 11.0 (TID 6) in 49 ms on 172.20.0.4 (executor 0) (1
/1)
26/09/15 16:11:04 INFO TaskSchedulerImpl: Removed TaskSet 11.0 whose tasks have all completed, from pool
26/09/15 16:11:04 INFO DAGScheduler: ResultStage 11 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:176
8) finished in 53 ms
26/09/15 16:11:04 INFO DAGScheduler: Job 6 is finished. Cancelling potential speculative or zombie tasks for this job
26/09/15 16:11:04 INFO TaskSchedulerImpl: Cancelling stage 11
26/09/15 16:11:04 INFO TaskSchedulerImpl: Killing all running tasks in stage 11: Stage finished
26/09/15 16:11:04 INFO DAGScheduler: Job 6 finished: $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:176
8, took 56.84775 ms
26/09/15 16:11:04 INFO DAGScheduler: Asked to cleanup jobs for query execution 2
34 words scanned
31 words of at least 3 characters
14 distinct words
26/09/15 16:11:04 INFO FileSourceStrategy: Pushed Filters:
26/09/15 16:11:04 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/09/15 16:11:04 INFO CodeGenerator: Code generated in 18.158208 ms
26/09/15 16:11:04 INFO MemoryStore: Block broadcast_10 stored as values in memory (estimated size 214.5 KiB, free 433
.5 MiB)
26/09/15 16:11:04 INFO MemoryStore: Block broadcast_10_piece0 stored as bytes in memory (estimated size 39.5 KiB, fre
e 433.5 MiB)
26/09/15 16:11:04 INFO SparkContext: Created broadcast 10 from $anonfun$withThreadLocalCaptured$1 at CompletableFuture
.java:1768
26/09/15 16:11:04 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, c
onsidered as scanning 4194304 bytes.
26/09/15 16:11:04 INFO DAGScheduler: Registering RDD 27 ($anonfun$withThreadLocalCaptured$1 at Comple
:1768) as input to shuffle 4
26/09/15 16:11:04 INFO DAGScheduler: Got map stage job 7 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.jav
```

```
main.py 4, M X
ITCS6190-H6-Hardi > main.py > ...
37 # ----- Step 0: load the data
38 # Declare the schemas up front (slides "Define Schema"): no inference pass over the file,
39 # and the timestamp column becomes a real timestamp instead of a string.
40 #
41 # TODO: write the schema of listening_logs.csv as a StructType with four fields:
42 # user_id STRING, song_id STRING, timestamp TIMESTAMP, duration_sec INT
43 logs_schema = StructType([
44     StructField("user_id", StringType(), True), You, 9 seconds ago · Uncommitted changes
45     StructField("song_id", StringType(), True),
46     StructField("timestamp", TimestampType(), True),
47     StructField("duration_sec", IntegerType(), True)
48 ])

Problems 4 Output Terminal Ports
zsh - ITCS6190-H6-Hardi

(base) hardikoradiya@hardis-MacBook-Air ITCS6190-H6-Hardi:~/input$ submit \
--master spark://spark-master:7077 \
/opt/spark/work-dir/main.py \
/opt/spark/work-dir/shared/input \ ...

26/09/22 22:17:09 INFO SharedState: Warehouse path is 'file:/opt/spark/work-dir/spark-warehouse'.
26/09/22 22:17:09 INFO InMemoryFileIndex: It took 14 ms to list leaf files for 1 paths.
26/09/22 22:17:10 INFO InMemoryFileIndex: It took 1 ms to list leaf files for 1 paths.
root
|-- user_id: string (nullable = true)
|-- song_id: string (nullable = true)
|-- timestamp: timestamp (nullable = true)
|-- duration_sec: integer (nullable = true)

26/09/22 22:17:10 INFO FileSourceStrategy: Pushed Filters:
26/09/22 22:17:10 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/09/22 22:17:10 INFO CodeGenerator: Code generated in 90.368708 ms
26/09/22 22:17:10 INFO MemoryStore: MemoryStore started with capacity 434.4 MiB
26/09/22 22:17:10 INFO MemoryStore: Block broadcast_0 stored as values in memory (estimated size 214.4 KiB, free 434.2 MiB)
26/09/22 22:17:10 INFO MemoryStore: Block broadcast_0_piece0 stored as bytes in memory (estimated size 39.5 KiB, free 434.2 MiB)
26/09/22 22:17:10 INFO SparkContext: Created broadcast 0 from $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768
```

HandsOnL6

main.py 4

ITCS6190-H6-Hardi > main.py

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Explorer

- HandsOnL6
 - ITCS6190-H6-Hardi
 - conf
 - shared-folder
 - input
 - .gitkeep
 - listening_logs.csv
 - songs_metadata.csv
 - output
 - .gitignore
 - datagen.py
 - docker-compose.codespac...
 - docker-compose.yml
 - main.py
 - README.md
 - REPORT.md

Outline

Timeline

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zsh - ITCS6190-H6-Hardi

```
(base) hardikoradiya@hardis-MacBook-Air HandsOnL6 % ls
ITCS6190-H6-Hardi
(base) hardikoradiya@hardis-MacBook-Air HandsOnL6 % cd ITCS6190-H6-Hardi
(base) hardikoradiya@hardis-MacBook-Air ITCS6190-H6-Hardi % docker compose ps
```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS
spark-master	apache/spark:4.2.0	"/opt/entrypoint.sh ..."	spark-master	35 seconds ago	Up 34 seconds
spark-worker-1	apache/spark:4.2.0	"/opt/entrypoint.sh ..."	spark-worker-1	35 seconds ago	Up 34 seconds
spark-worker-2	apache/spark:4.2.0	"/opt/entrypoint.sh ..."	spark-worker-2	35 seconds ago	Up 34 seconds

1 file and 0 cells to analyze

UTF-8 LF {} Python Python 3.9.6.final.0 Go Live Antigravity - Settings Prettier

HandsOnL6

main.py 4

ITCS6190-H6-Hardi > main.py

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Explorer

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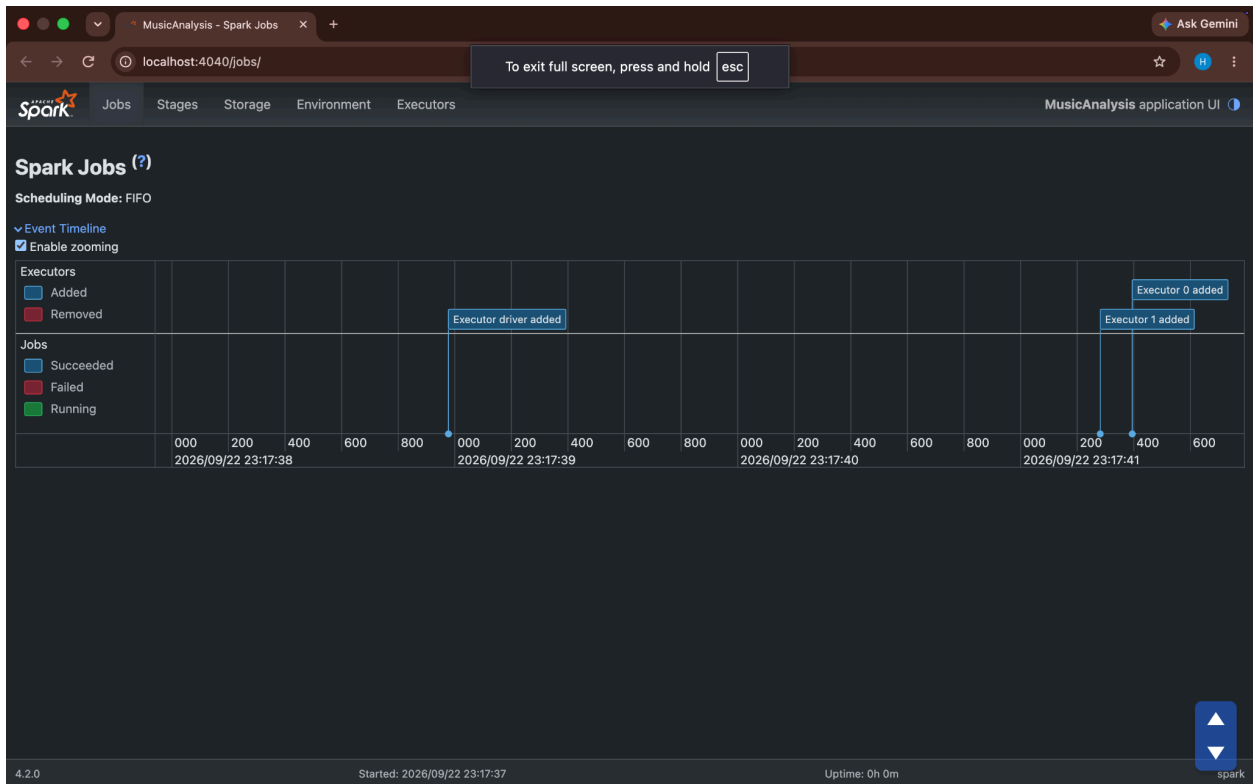
Ports

zsh - ITCS6190-H6-Hardi

```
(base) hardikoradiya@hardis-MacBook-Air ITCS6190-H6-Hardi % python3 datagen.py 801484363
ardi/shared-folder/input
(base) hardikoradiya@hardis-MacBook-Air ITCS6190-H6-Hardi % head shared-folder/input/listening_logs.csv
user_id,song_id,timestamp,duration_sec
user_91,song_7,2026-03-13 19:34:24,66
user_66,song_14,2026-03-12 14:43:50,77
user_24,song_35,2026-03-24 22:04:51,243
user_10,song_31,2026-03-17 15:53:04,65
user_13,song_32,2026-03-24 23:52:47,138
user_40,song_11,2026-03-15 06:13:49,151
user_72,song_16,2026-03-20 06:24:20,173
user_45,song_3,2026-03-19 10:26:38,221
user_79,song_24,2026-03-14 01:07:08,96
(base) hardikoradiya@hardis-MacBook-Air ITCS6190-H6-Hardi % head shared-folder/input/songs_metadata.csv
song_id,title,artist,genre,mood
song_1,Title_song_1,Artist_4,Hip-Hop,Chill
song_2,Title_song_2,Artist_13,Jazz,Chill
song_3,Title_song_3,Artist_4,Hip-Hop,Energetic
song_4,Title_song_4,Artist_18,Hip-Hop,Happy
song_5,Title_song_5,Artist_14,Pop,Energetic
song_6,Title_song_6,Artist_3,Jazz,Energetic
song_7,Title_song_7,Artist_6,Jazz,Sad
song_8,Title_song_8,Artist_11,Rock,Chill
song_9,Title_song_9,Artist_13,Jazz,Sad
(base) hardikoradiya@hardis-MacBook-Air ITCS6190-H6-Hardi %
```

1 file and 0 cells to analyze

UTF-8 LF {} Python Python 3.9.6.final.0 Go Live Antigravity - Settings Prettier



MusicAnalysis - Stages for All

localhost:4040/stages/

MusicAnalysis application UI

Stages for All Jobs

Active Stages: 1
Pending Stages: 1
Completed Stages: 1

Active Stages (1)

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Stage Id	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
2	<code>\$anonfun\$withThreadLocalCaptured\$1 at CompletableFuture.java:1768</code> +details (kill)	2026/09/22 23:18:13	Unknown	0/1				

Pending Stages (1)

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Stage Id	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
1	<code>\$anonfun\$withThreadLocalCaptured\$1 at CompletableFuture.java:1768</code> +details	Unknown	Unknown	0/1				

Completed Stages (1)

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Stage Id	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
0	<code>\$anonfun\$withThreadLocalCaptured\$1 at CompletableFuture.java:1768</code> +details	2026/09/22 23:18:13	0.7 s	1/1	39.5 KiB			59.0 B

MusicAnalysis - Details for Stage 0

localhost:4040/stages/stage/?id=0&attempt=0

Ask Gemini

JobsStagesStorageEnvironmentExecutorsSQL / DataFrame

MusicAnalysis application UI

Details for Stage 0 (Attempt 0)

Submitted: 2026/09/22 23:18:13
Duration 0.7 s
Resource Profile Id: 0
Total Time Across All Tasks: 0.3 s
Locality Level Summary: Process local: 1
Input Size / Records: 39.5 KiB / 1000
Shuffle Write Size / Records: 59.0 B / 1
Associated Job Ids: 0

> DAG Visualization
> Show Additional Metrics
> Event Timeline

Summary Metrics for 1 Completed Tasks

Metric	Min	25th percentile	Median	75th percentile	Max
Duration	0.3 s	0.3 s	0.3 s	0.3 s	0.3 s
GC Time	6.0 ms	6.0 ms	6.0 ms	6.0 ms	6.0 ms
Input Size / Records	39.5 KiB / 1000	39.5 KiB / 1000	39.5 KiB / 1000	39.5 KiB / 1000	39.5 KiB / 1000
Shuffle Write Size / Records	59 B / 1	59 B / 1	59 B / 1	59 B / 1	59 B / 1

> Aggregated Metrics by Executor

Tasks (1)

Show 20 entries

Search:

Index	Task ID	Attempt	Status	Locality level	Executor ID	Host	Logs	Launch Time	Duration	GC Time	Shuffle Read Fetch Wait Time	Shuffle Remote Reads	Input Size / Records	Shuffle Write Size / Records	Error
0	0	0	SUCCESS	PROCESS_LOCAL	0	172.20.0.4	stdout	2026-09-22	0.3 s	6.0 ms	0.0 ms	0.0 B	39.5 KiB /	59 B / 1	